# PGRR: Planning-Guided Failure-Triggered Recovery and Rejoin for Dynamic Social Navigation

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***Abstract*—Classical navigation is dependable in routine mapped environments but can freeze, oscillate, or collide in reciprocal pedestrian interactions. We present PGRR, a bounded recovery layer that leaves Nav2 DWB in charge of nominal motion and invokes learning only after an observable failure trigger. A compact policy selects among 21 temporary subgoals and four interpretable modes; a planning mask rejects infeasible actions and an independent supervisor may stop unsafe motion. Training uses a privileged short-horizon rollout expert, behavior cloning, and DAgger, whereas deployment uses only LiDAR and navigation history. A locked Gazebo study contains 64 logical episodes across eight interaction families and three densities. In the 24 paired Base–PGRR tests, Base and PGRR achieve success rates of 20.8% and 33.3%, and collision rates of 79.2% and 0.0%, respectively. The paired success-rate difference is +12.5 pp (95% bootstrap interval [+0.0 pp, +29.2 pp], Holm-adjusted $p = 0.750$). The results quantify a condition-dependent safety–completion trade-off rather than universal superiority. All raw streams, exclusions, statistics, figures, and manuscript numbers are generated under a frozen manifest.**

## I. Introduction

Classical navigation stacks remain attractive because mapping, global planning, local collision avoidance, and safety checks are explicit and independently testable. Their local planners can nevertheless freeze, oscillate, or enter reciprocal deadlock when moving pedestrians block a narrow passage. End-to-end learning can encode interaction patterns, but replacing the nominal controller also changes behavior in routine states where the classical stack is already competent.

We study a narrower question: can learning be confined to the short interval in which a classical planner is failing? Our system, PGRR (Planning-Guided Recovery and Rejoin), leaves Nav2 DWB in control during normal PointGoal navigation. An observable rule detector and a hysteretic state machine invoke a recovery policy only for collision risk, freeze, oscillation, or deadlock. The policy does not command wheel velocity. It selects one of 21 local temporary goals or WAIT, BACKUP, REPLAN, and CONTINUE; the classical stack executes the goal and PGRR then reissues the original task goal. A planning-derived action mask and an independent braking supervisor bound every learned decision.

Training uses information that is deliberately unavailable at test time. A short-horizon expert rolls out each legal recovery action using the map and true simulated pedestrian state. Behavior cloning initializes a compact policy, and DAgger relabels states visited by the learned controller. Deployment receives only LiDAR history, the local path and goal, robot and planner commands, progress history, planner status, and rule-detector scores.

The contributions are:

- a failure-triggered ROS2 hierarchy that preserves the authority of a classical planner outside explicitly detected recovery intervals;
- a fixed, interpretable recovery action space with planning and sensor masks enforced during expert labeling and online policy execution;
- an automatically labeled BC–DAgger pipeline for learner-induced recovery states, together with an offline ablation of DAgger and action masking; and
- a reproducible evaluation over eight pedestrian-interaction families, three densities, four recovery mechanisms, immutable raw streams, and paired statistical analysis.

We claim neither formal collision avoidance nor a general solution to social navigation. The experiments quantify when bounded intervention helps, when its conservatism increases travel time, and which failure families remain unresolved.

## II. Related Work

### A. Classical and Social Navigation

The dynamic-window approach searches dynamically feasible velocity commands over a short horizon [1]; Nav2 provides a modern ROS2 navigation system in which DWB implements this family of local control [2]. Reciprocal methods such as ORCA [3] and learned crowd-aware policies such as SARL [4] instead model interaction more directly. The broader social-navigation literature emphasizes that collision avoidance, comfort, legibility, and task efficiency are distinct objectives [5]. PGRR retains DWB as the nominal controller and evaluates geometric clearance and discomfort time; it does not claim to model all social norms.

### B. Hybrid Planning and Recovery

All-in-One learns a continuous switch among complete navigation planners [6], whereas DR-MPC learns a residual around model-predictive control [7]. Both demonstrate the value of hybrid model-based and learned navigation. PGRR differs in authority and output: intervention is failure-triggered, and the learned output is a temporary subgoal or an interpretable recovery mode rather than a nominal planner choice or velocity residual.

Navigation recovery itself is not new. Del Duchetto *et al.* learn local failure detection and recovery from human

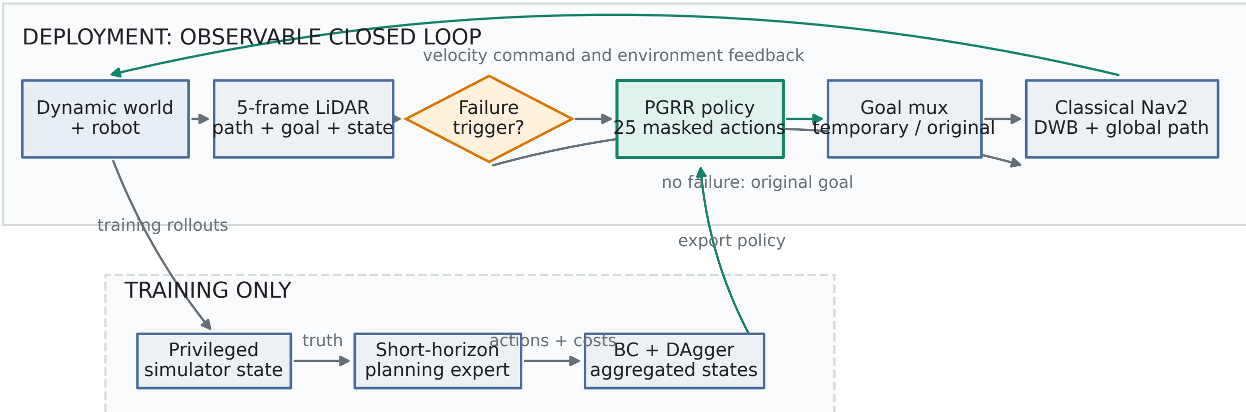


Fig. 1. PGRR architecture. Privileged simulator state and expert rollouts are confined to training (dashed region); deployment observes LiDAR and navigation history. Nav2 retains nominal and temporary-goal execution.

demonstrations [8]; Mohammad *et al.* predict planning failure and search for a recovery state using a Gaussian-process model [9]. Our focus is automatic privileged-planner labels, planning-mask enforcement, and aggregation on states induced by the recovery learner. DAgger supplies the standard reduction for this sequential distribution shift [10]. Invalid-action masking has also been analyzed in policy-gradient learning [11]; here it is used as a deterministic planning constraint for imitation inference, not as a safety proof.

### C. Simulation and Evaluation

Arena-Bench introduced reproducible dynamic-obstacle evaluation [12], and later Arena releases expanded social-navigation scenarios and ROS2 support [13], [14], [15]. Our installed, pinned Humble profile is an earlier Gazebo fallback rather than Arena 5.0. We therefore use the later releases only as platform context and fully disclose the deterministic pedestrian proxy used by the experiments.

## III. Problem Formulation and Method

### A. Problem and Observable State

Let a differential-drive robot navigate a known static occupancy map from pose $x_0$ to goal $g$ while a classical planner $B$ supplies command $u_t^B$. PGRR must attempt recovery without using pedestrian truth at deployment. Its history $o_t$ contains five 180-beam LiDAR scans, goal distance and bearing, eight path waypoints in the robot frame, robot velocity, $u_t^B$, ten-step goal-distance and angular-velocity histories, a one-hot planner status, and four rule scores. Ground-truth pedestrian state is isolated in a privileged record used only for expert labeling and evaluation.

### B. Failure Trigger and Recovery State Machine

The detector computes collision-risk, freeze, oscillation, and deadlock scores. Collision risk combines braking distance, 1.5-s time to collision, absolute near-field clearance, and a 0.5-s radial closing trend. Freeze requires more than 1 m remaining, less than 0.15 m displacement over 3 s, and a planner request for motion. Oscillation requires at least six angular-velocity sign changes after a 0.08 rad/s deadband and less than 0.20 m progress over 4 s. Deadlock uses less than 0.15 m displacement and less than 0.15 m goal progress over 4 s, an obstacle within 1.2 m, and low robot speed. Motion rules are disabled for the first 4 s after the synchronized episode handshake; collision evidence remains active.

The usual state sequence is NORMAL→PENDING→RECOVERY→REJOIN, although a confirmed one-frame trigger may enter RECOVERY directly after cooldown. EMERGENCY-STOP has highest priority and may resume; only FAILED and SUCCEEDED are terminal. Recovery enters above $\tau_{on} = 0.65$ and exits after four frames below $\tau_{off} = 0.35$ plus valid goal progress. Decisions occur at 2 Hz, velocity control at 10 Hz, with a 0.5-s minimum hold, 2-s cooldown, an 8-s learned-policy recovery bound, a 5-s rejoin bound, and four consecutive recovery attempts. A 30-s extension exists only for the privileged online Oracle's yield option and is not used by PGRR. The original goal is stored before intervention and reissued whenever a bounded temporary option hands control to REJOIN; this does not imply that Nav2 reached the temporary target.

### C. Recovery Actions, Mask, and Safety Supervisor

The 25 actions are 21 robot-relative subgoals

$$(r, \theta) \in \{0.6, 1.0, 1.4\}\,\text{m} \times \{-90, -60, -30, 0, 30, 60, 90\}^\circ, \quad (1)$$

and WAIT, BACKUP, REPLAN, and CONTINUE. For action $a$, a Boolean mask $M(o_t, a)$ rejects occupied or off-map endpoints, disconnected grid components, an obstructed swept LiDAR segment, insufficient rear observability, excessive global-path deviation, or unavailable replanning. WAIT is the final safe fallback, so at least one action remains legal. Before softmax, masked logits are set to the minimum representable value.

An independent supervisor may override any policy output. Its longitudinal stopping distance is

$$d_{stop} = \frac{v^2}{2a_b} + v\ell + m, \tag{2}$$

with $a_b = 0.8$ m/s$^2$, latency $\ell = 0.15$ s, and margin $m = 0.45$ m in the recovery controller. It provides bounded stop, reverse, turn, and forward escape submodes, all logged separately from the learned action. This is an empirical safety filter, not a formal guarantee.

### *D. Privileged Planning Expert*

At training time the expert constructs local occupancy from the recorded LiDAR and observes privileged robot and pedestrian positions. Pedestrian velocities are estimated by finite differences rather than read as simulator truth. It uses constant-velocity prediction with conservative yielding and uncertainty inflation, then simulates every legal action for 3 s at 0.1-s resolution. Subgoal rollouts use grid A* and pure pursuit with differential-drive kinematics; the four mode actions have matched short-horizon models. The cost is

$$J(a) = w_c J_c + w_p J_p + w_s J_s + w_r J_r + w_l J_l \tag{3}$$
$$+ w_{sm} J_{sm} + w_t J_t + w_{sw} J_{sw} + w_w J_w, \tag{4}$$

where the terms denote collision, insufficient progress, social-distance integral, path-rejoin distance, length, angular smoothness, time, side switching, and repeated waiting. The weights are respectively $(10^6, 8, 4, 3, 0.4, 0.2, 0.3, 0.5, 1)$; collision and masked actions are effectively hard constraints. Besides the minimum-cost action, the expert stores all 25 costs and the gap to the second-best action for analysis.

### *E. Behavior Cloning and DAgger*

The policy uses two 1-D convolutions over the $5 \times 180$ LiDAR stack, a 128-unit state encoder, and a two-layer head, totaling 59,193 parameters. For the selected model, uniform behavior cloning minimizes

$$\mathcal{L}_{BC} = -\frac{1}{N} \sum_i \log \pi_\theta(a_i^* \mid o_i, M_i). \tag{5}$$

DAgger then executes the learner on training scenarios, asks the privileged expert to relabel visited recovery states, aggregates them, and retrains. The selected aggregate contains 1,387 raw recovery states from 12 episodes; mirror augmentation yields 2,774 training samples, with 399 states in a scenario-disjoint validation set. Training uses AdamW at $3\times10^{-4}$, batch size 128, maximum 30 epochs, gradient norm 1, and early-stopping patience 6. The deployment checkpoint is selected by validation loss. Its configuration has zero margin weight and no PPO; accordingly neither MWBC nor reinforcement learning is claimed as part of PGRR.

## IV. Experimental Setup

### *A. Platform and Robot*

Experiments run in a pinned ROS2 Humble Arena Docker image (Arena commit `c2ff4a87`) with Gazebo, Nav2 NavFn/DWB, and a Jackal model. Flatland is not available in this profile. Scenario-coordinate bounds span $31.28 \times 24.03$ m. The evaluation footprint radius is 0.36 m; the loaded DWB limits are 0.26 m/s and 1.0 rad/s. A 360-degree, 360-sample planar GPU LiDAR runs at 10 Hz and is resampled to 180 bins. Pose-derived simulation odometry supplies the declared known-pose interface, while evaluation independently uses Gazebo model poses for physical goal and contact checks.

The installed profile lacks the complete HuNav chain. We therefore use deterministic, LiDAR-visible, contactless cylindrical pedestrians following scenario routes. Most actors hold when a candidate step would both enter a 1.3 m robot-clearance region and reduce separation; temporary blockage uses a finite route and a 0.8 m threshold. This removes rigid-body impulses but is less realistic than a responsive crowd model; conclusions are restricted to dynamic pedestrian navigation in simulation.

### *B. Scenarios, Splits, and Locked Manifest*

Eight families exercise distinct local failures: head-on corridor, doorway bottleneck, crossing flow, blind corner, group blocking, overtaking, opposite streams, and temporary blockage. Low, medium, and high densities contain 2, 4, and 6 pedestrians, with family-specific speed ranges of 0.08–0.75 m/s. Train, validation, and test use disjoint compiled scenario variants and seed ranges 1000, 2000, and 3000, respectively.

Before test execution we tagged commit `35d7e60` as `pre-final-eval`, hashed the scenario manifest, checkpoint, detector, state-machine, and action configurations, and prohibited further tuning. The primary paired test runs Base DWB and PGRR on all eight families and three densities (48 episodes). At high density, Standard Nav2 Recovery and the rule-triggered Heuristic hierarchy add 16 episodes, for 64 algorithm episodes total. Each family–density cell has one fixed test seed; this broad coverage does not substitute for multi-seed replication and is reported as a limitation.

### *C. Baselines and Ablations*

The evaluated closed-loop methods are: (i) Base DWB with periodic global replanning and no recovery subtree; (ii) Arena's Standard Nav2 recovery behavior tree; (iii) the same failure trigger, state machine, action mask, and safety supervisor with deterministic Heuristic action selection; and (iv) PGRR with the selected DAgger policy. The primary pair isolates addition of the complete hierarchy to Base. The high-density four-method subset separates standard, heuristic, and learned recovery mechanisms.

The training workflow completed two aggregation rounds on the train split. Validation rejected the second-round checkpoint; the frozen model instead extends the better first-round aggregate with one train-only head-on coverage shard. Its selected

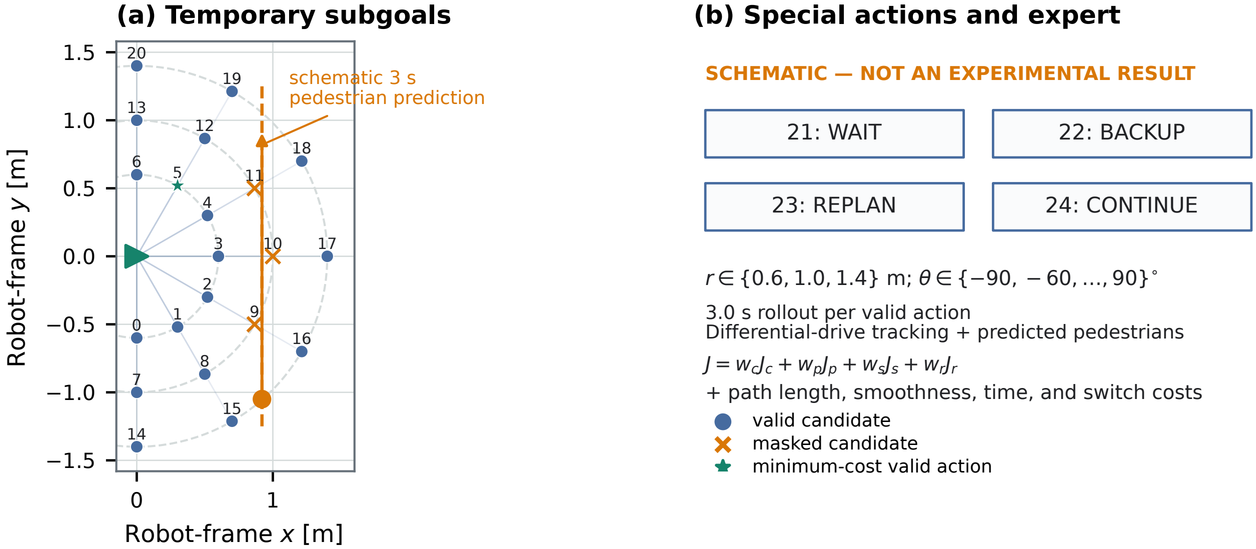


Fig. 2. Interpretable recovery decisions and privileged expert labeling. Candidate rollouts that violate the configured planning mask are never selectable; privileged pedestrian prediction is used only to rank legal training actions.

aggregate contains 1,387 recovery states from 12 episodes, and no test episode entered model selection.

A separate offline ablation evaluates Uniform BC, Margin-Weighted BC, and the selected DAgger checkpoint on the same 399 scenario-disjoint recovery states. Each model is evaluated both with the original mask and with an all-legal mask; the latter reports unsafe proposals only and is never executed in simulation.

### D. Metrics and Statistical Protocol

Every 10-Hz stream records robot and pedestrian geometry, commands, planner and recovery state, failure scores, LiDAR, and paths. We report terminal success, collision, timeout, planner failure, SPL, physical path length, navigation time, minimum human-centre distance, time below 1.0 m, fraction below 1.2 m, angular jerk, emergency stops, triggers, recovery duration, and intervention ratio. `SIMULATOR_FAILURE` and `INVALID_RESET` attempts are retained, counted, and retried once, but excluded from algorithm metrics. All other failures remain.

Base and PGRR are paired by the same immutable scenario and seed. Binary outcomes use exact McNemar tests; continuous paired differences use two-sided Wilcoxon signed-rank tests. Mean differences receive 10,000-sample paired bootstrap 95% confidence intervals, and related $p$-values use Holm correction. We report effects and confidence intervals even when significance is absent.

### E. Execution Integrity

Concurrent workers use distinct ROS domain IDs and Gazebo partitions. A start handshake holds pedestrians until navigation and logging are active. Goal reach requires a physical-pose confirmation; supported shelves use a robot-circle-to-oriented-box distance test. Near-pedestrian LiDAR consistency is evaluation-only. The runner refuses overwrite, preserves both physical attempts after an infrastructure retry, and fails unless each logical manifest row has exactly one valid algorithm outcome. Tables and figures are regenerated from the resulting Parquet/JSON artifacts rather than hand-entered values.

## V. Results and Analysis

### A. Closed-Loop Navigation Outcomes

Table I reports every retained algorithm outcome. Across the 24 matched family–density conditions, Base DWB reaches 20.8% of goals, collides in 79.2%, and times out in 0.0%. PGRR reaches 33.3%, collides in 0.0%, and times out in 66.7%. Its mean SPL is 0.303 versus 0.205 for Base. Figure 4 retains the density and family structure that a pooled rate can hide.

The paired PGRR-minus-Base success difference is +12.5 pp with a 95% bootstrap interval [+0.0 pp, +29.2 pp]; its exact McNemar test gives Holm-adjusted $p = 0.750$. The collision difference is −79.2 pp [−91.7 pp, −62.5 pp] ($p_{\mathrm{Holm}}$ =< 0.001), and the timeout difference is +66.7 pp [+45.8 pp, +83.3 pp] ($p_{\mathrm{Holm}}$ =< 0.001). Table II reports these effects together with the continuous paired tests. We interpret intervals and discordant pairs rather than using an unadjusted $p$-value as a binary performance verdict.

### B. Safety, Efficiency, and Intervention

Mean minimum human-centre distance is 1.318 m for Base and 1.267 m for PGRR. Their paired difference is −0.051 m [−0.498 m, +0.251 m] with $p_{\mathrm{Holm}} = 0.257$. Among episodes where both methods reach the goal, the mean PGRR-minus-Base navigation-time difference is +9.4 s [−2.1 s, +30.8 s] ($p_{\mathrm{Holm}} = 1.000$). The raw successful-run means are 85.6 s

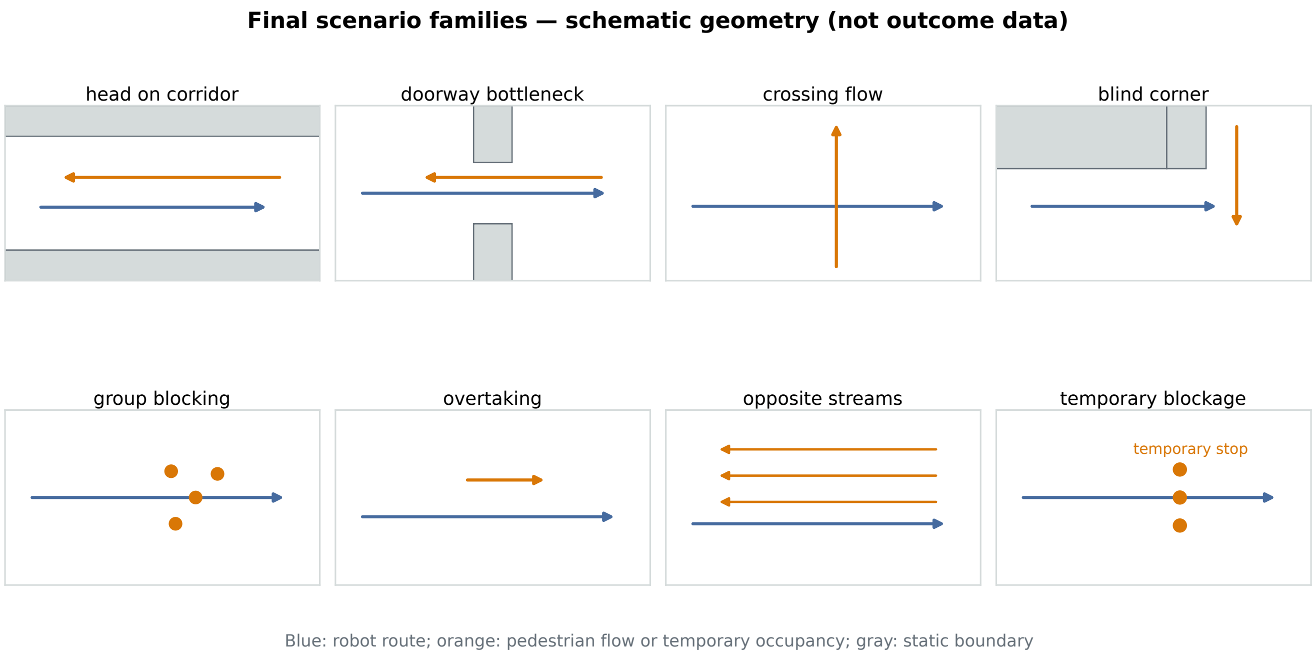


Fig. 3. The eight locked test families. Each panel is generated from its checked-in scenario definition and shows the static layout, robot start and goal, reference route, and initial pedestrian motion.

TABLE I

LOCKED FINAL EVALUATION. DWB AND PGRR USE THE FULL 24-EPISODE MANIFEST; STANDARD AND HEURISTIC RECOVERY USE THE PAIRED 8-EPISODE HIGH-DENSITY SUBSET. RATES USE VALID ALGORITHM EPISODES, AND NAVIGATION TIME IS REPORTED ONLY FOR SUCCESSFUL EPISODES. VALUES AFTER ± ARE STANDARD DEVIATIONS. "EXCL." COUNTS RETAINED SIMULATOR-FAILURE OR INVALID-RESET PHYSICAL ATTEMPTS THAT WERE RETRIED AND EXCLUDED FROM ALGORITHM RATES.

| Method | $N$ | Success [%] | Collision [%] | Timeout [%] | SPL | Time [s] | $d_{\min}$ [m] | Excl. |
|---|---|---|---|---|---|---|---|---|
| DWB | 24 | 20.8 | 79.2 | 0.0 | 0.205 ± 0.409 | 85.6 ± 10.7 | 1.318 ± 1.311 | 1 |
| Standard | 8 | 12.5 | 87.5 | 0.0 | 0.106 ± 0.299 | 114.5 | 1.008 ± 0.949 | 0 |
| Heuristic | 8 | 12.5 | 0.0 | 87.5 | 0.108 ± 0.306 | 129.4 | 1.092 ± 0.306 | 0 |
| PGRR | 24 | 33.3 | 0.0 | 66.7 | 0.303 ± 0.440 | 111.9 ± 32.2 | 1.267 ± 0.581 | 2 |

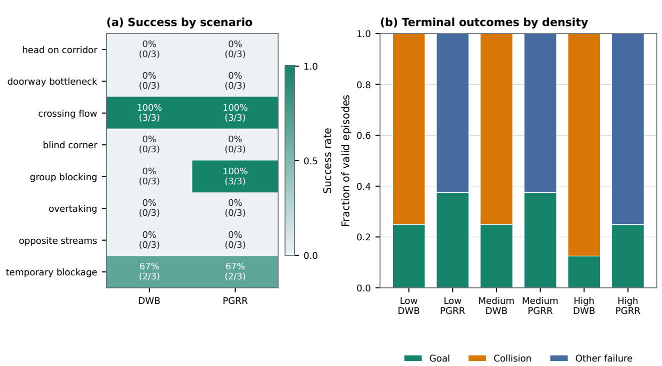


Fig. 4. Terminal outcomes by density and scenario family. Each family–density cell is one locked test seed; the plot exposes mixed effects and must not be read as a per-family probability estimate.

and 111.9 s. Figure 5 compares all four methods on their common high-density subset, while Table III shows the resulting clearance–time trade-off, emergency interventions, recovery triggers, and rejoin success. A timeout that avoids contact remains a timeout, not a recovery success. The paired tests also show that PGRR uses more emergency stops, spends more time under intervention, and has higher angular jerk. Its mean personal-space violation ratio increases, while neither minimum human distance nor discomfort time remains different after Holm correction. Thus zero observed contacts in this manifest must not be conflated with smoother or uniformly more social motion.

### C. Recovery Dynamics and Run Evidence

Figure 6 aligns failure score, goal progress, recovery state, and selected actions for a representative final episode chosen by a deterministic artifact rule. Figure 7 reconstructs three instants from that episode's 10-Hz telemetry: robot and pedestrian poses, the global path, and recovery state. It is explicitly a telemetry reconstruction rather than a simulator camera image. Together they verify that the policy intervenes transiently and that the original goal is restored; they are mechanism evidence, not additional trials.

TABLE II

PAIRED STATISTICAL COMPARISONS FROM THE LOCKED FINAL MANIFEST. CONFIDENCE INTERVALS ARE 95%; $p_{\text{Holm}}$ DENOTES MULTIPLICITY-CORRECTED VALUES.

| Comparison | Metric | Test | Estimate [95% CI] | $p$ | $p_{\text{Holm}}$ | Effect | $N_{\text{pairs}}$ |
|---|---|---|---|---|---|---|---|
| DWB vs PGRR | Success rate | McNemar exact | 0.125 [0.000, 0.292] | 0.250 | 0.750 | 7.000 | 24 |
| DWB vs PGRR | Collision rate | McNemar exact | -0.792 [-0.917, -0.625] | < 0.001 | < 0.001 | 0.026 | 24 |
| DWB vs PGRR | Timeout rate | McNemar exact | 0.667 [0.458, 0.833] | < 0.001 | < 0.001 | 33.000 | 24 |
| DWB vs PGRR | Successful time [s] | Wilcoxon signed-rank | 9.431 [-2.125, 30.776] | 1.000 | 1.000 | 0.067 | 5 |
| DWB vs PGRR | Successful path length [m] | Wilcoxon signed-rank | 0.633 [-0.439, 2.360] | 1.000 | 1.000 | -0.067 | 5 |
| DWB vs PGRR | SPL | Wilcoxon signed-rank | 0.098 [-0.004, 0.222] | 0.173 | 0.692 | 0.619 | 24 |
| DWB vs PGRR | Minimum human distance [m] | Wilcoxon signed-rank | -0.051 [-0.498, 0.251] | 0.037 | 0.257 | 0.487 | 24 |
| DWB vs PGRR | Personal-space violation | Wilcoxon signed-rank | 0.051 [0.002, 0.102] | 0.058 | 0.349 | 0.509 | 24 |
| DWB vs PGRR | Discomfort time [s] | Wilcoxon signed-rank | -0.690 [-2.360, 1.131] | 0.135 | 0.677 | -0.412 | 24 |
| DWB vs PGRR | Emergency stops | Wilcoxon signed-rank | 14.333 [10.792, 17.792] | < 0.001 | < 0.001 | 1.000 | 24 |
| DWB vs PGRR | Recovery triggers | Wilcoxon signed-rank | 5.208 [3.917, 6.583] | < 0.001 | < 0.001 | 1.000 | 24 |
| DWB vs PGRR | Intervention ratio | Wilcoxon signed-rank | 0.580 [0.462, 0.688] | < 0.001 | < 0.001 | 0.987 | 24 |
| DWB vs PGRR | Mean angular jerk | Wilcoxon signed-rank | 0.575 [0.407, 0.748] | < 0.001 | < 0.001 | 0.980 | 24 |

TABLE III

RECOVERY BEHAVIOR ON VALID FINAL EPISODES. ENTRIES ARE EPISODE-LEVEL MEAN ± STANDARD DEVIATION; UNDEFINED RECOVERY RATES FOR METHODS WITHOUT A TRIGGER ARE SHOWN AS DASHES.

| Method | $N$ | Triggers / ep. | Recovery success [%] | Duration [s] | Intervention [%] | Emergency stops / ep. |
|---|---|---|---|---|---|---|
| DWB | 24 | 0.00 ± 0.00 | – | 0.00 ± 0.00 | 4.0 ± 3.9 | 0.00 ± 0.00 |
| Standard | 8 | 0.00 ± 0.00 | – | 0.00 ± 0.00 | 5.0 ± 4.0 | 0.00 ± 0.00 |
| Heuristic | 8 | 6.12 ± 4.05 | 62.9 ± 31.6 | 132.53 ± 40.62 | 75.7 ± 19.6 | 14.12 ± 8.08 |
| PGRR | 24 | 5.21 ± 3.45 | 73.6 ± 15.3 | 105.28 ± 58.42 | 61.9 ± 30.0 | 14.33 ± 8.86 |

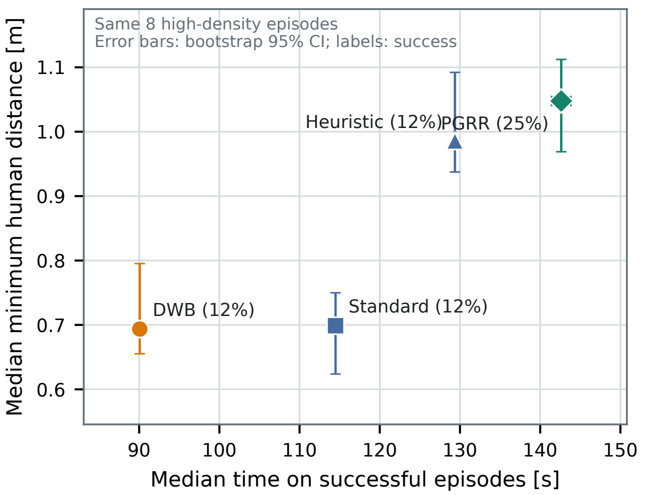


Fig. 5. Safety–efficiency view on the common eight-condition high-density subset. Marker shape encodes terminal outcome, preventing long collision-free stagnation from appearing as task success.

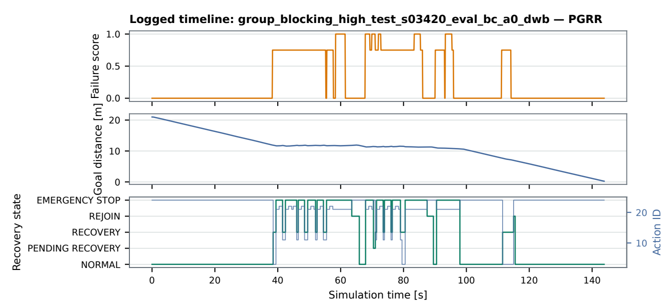


Fig. 6. Recorded failure score, goal distance, and recovery state for a representative final episode. All event times come from the raw JSONL stream.

## D. Imitation and Mask Ablations

Table IV compares Uniform BC, Margin-Weighted BC, and DAgger on the same 399 scenario-disjoint validation states. In this dataset, margin weighting does not change the reported decision metrics and is therefore not part of the selected method. DAgger reduces expert-cost regret and improves top-$k$ agreement. Disabling the mask produces many invalid proposals, so the no-mask rows are an offline diagnostic and were never executed on the robot. These results support distribution aggregation and mask enforcement, but they do not isolate the online safety supervisor from the learned policy.

## E. Failure Cases

PGRR reaches every crossing-flow and group-blocking goal, plus two of three temporary-blockage goals, but times out in all head-on-corridor, doorway, and blind-corner conditions. The generated failure report classifies nine PGRR timeouts as terminal stagnation and seven as still making short-window progress; the latter group includes all blind-corner timeouts. The Heuristic avoids contact on the high-density subset but stagnates in seven of eight episodes. By contrast, Base and Standard usually terminate earlier by human or static contact. These terminal categories support a conservative-live-lock diagnosis, not attribution to a particular learned action: the final result table does not contain action-specific WAIT/BACKUP counts. All outcomes remain in the reported rates, motivat-

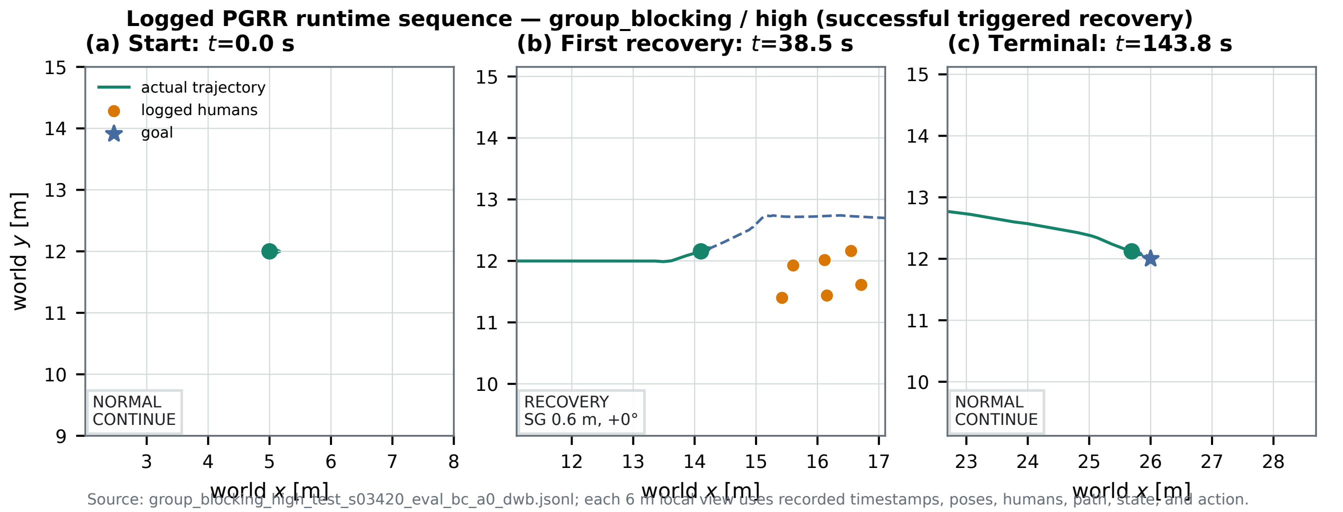


Fig. 7. Telemetry reconstruction at episode start, first recovery intervention, and termination. Circles are recorded physical pedestrian centres; the robot pose, path, action, and state are taken from the same raw stream.

TABLE IV
OFFLINE POLICY ABLATION ON THE SAME 399 HELD-OUT VALIDATION STATES. REGRET USES THE EXPERT COST, WHOSE HARD INVALID-ACTION PENALTY DOMINATES MASK-DISABLED ROWS. † MASK DISABLED DENOTES COUNTERFACTUAL OFFLINE PROPOSALS ONLY; THOSE ACTIONS WERE NEVER EXECUTED BY THE ROBOT OR SIMULATOR IN CLOSED LOOP.

| Model | Planning mask | Top-1 [%] | Top-3 [%] | Invalid [%] | Cost regret | Near-opt. [%] | Catastrophic [%] |
|---|---|---|---|---|---|---|---|
| Uniform BC | Enabled | 88.5 | 95.7 | 0.0 | 0.271 | 88.7 | 0.0 |
| Uniform BC | Disabled† | 0.0 | 91.0 | 93.2 | $9.32 \times 10^5$ | 0.0 | 93.2 |
| Margin-weighted BC | Enabled | 88.5 | 95.7 | 0.0 | 0.271 | 88.7 | 0.0 |
| Margin-weighted BC | Disabled† | 0.0 | 91.7 | 93.2 | $9.32 \times 10^5$ | 0.0 | 93.2 |
| Triggered DAgger | Enabled | 92.2 | 98.5 | 0.0 | 0.031 | 93.2 | 0.0 |
| Triggered DAgger | Disabled† | 12.0 | 41.9 | 84.0 | $8.40 \times 10^5$ | 12.8 | 84.0 |

ing longer-horizon reciprocal reasoning rather than post-test parameter changes.

## VI. LIMITATIONS

The study has several important limits. First, the sensing interface is planar LiDAR on a known two-dimensional map, and simulation localization is more accurate than a deployed AMCL stack. Second, the Gazebo fallback pedestrians follow deterministic routes and use a simple yielding rule; they do not capture human intent, group norms, perception uncertainty, or the diversity of real crowds. Minimum distance and discomfort time are therefore geometric proxy metrics, not validated human-comfort judgments.

Third, the privileged expert depends on simulator truth and constant-velocity prediction. Its 3-s horizon can miss long reciprocal interactions. The deployed action set is finite, and a conservative mask or emergency latch can turn a collision into a timeout rather than a successful recovery. The safety supervisor has no formal invariance or collision-avoidance proof. The selected offline labeler and Heuristic baseline also retain a legacy 270-degree angular calibration when interpreting 180 bins that actually span the 360-degree sensor. The online learned-policy mask uses the scan message's angular metadata, and the CNN sees the same full-scan resampling in training and test, but expert action costs can inherit this angular distortion. We retain the locked results and disclose the mismatch rather than correcting it after examining the test set. Fourth, each test family–density cell has one fixed seed; the 24 primary pairs support a cross-condition paired analysis but not a low-variance per-scenario success estimate. Finally, there is no real-robot study, photorealistic Arena 5 run, learned failure predictor, or PPO refinement in the selected system. These are deliberate claim boundaries rather than omitted positive results.

## VII. CONCLUSION

PGRR confines learning to a bounded navigation-recovery interface: classical DWB controls nominal motion, observable rules trigger intervention, and a planning mask restricts a compact DAgger policy to interpretable temporary goals and modes. A privileged short-horizon planner creates labels automatically, and planning masks are enforced during labeling and online inference. The locked evaluation and offline ablations distinguish collision avoidance from task completion and expose the cost of conservative intervention. The resulting evidence supports PGRR as a reproducible recovery architecture, not a universal social-navigation solution. More realistic pedestrians, multi-seed replication, longer-horizon expert reasoning, and hardware tests are the highest-value next steps.

## References


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